

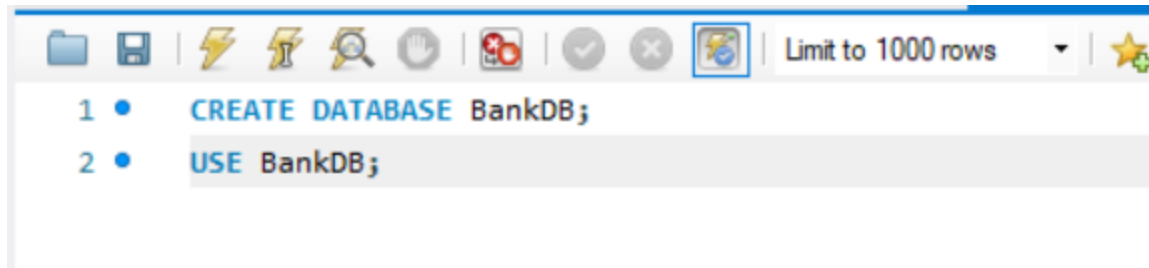
Database System Engineering and Distributed Backend Development

Course Code: 25CS13002E

ROLL NO:2520030597

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Query 1: -CREATE DATABASE



Output: -

#	Time	Action	Message	Duration / Fetch
✓ 1	14:24:44	CREATE DATABASE BankDB	1 row(s) affected	0.031 sec
✓ 2	14:24:47	USE BankDB	0 row(s) affected	0.000 sec

Query 2: - CREATE CUSTOMER TABLE

```
• CREATE TABLE Customers (  
    Customer_ID INT PRIMARY KEY,  
    Customer_Name VARCHAR(100) NOT NULL,  
    Phone VARCHAR(15),  
    Email VARCHAR(100),  
    City VARCHAR(50)  
);
```

Output: -

✓ 14	14:37:28	CREATE TABLE Customers (Customer_ID INT PRIMARY KEY, Customer_Name VARCHAR(100) NOT N...	0 row(s) affected
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Query 3: - CREATE ACCOUNT TABLE

```

    );
CREATE TABLE Account (
    Account_No INT PRIMARY KEY,
    Customer_ID INT,
    Account_Type VARCHAR(20),
    Balance DECIMAL(12,2) DEFAULT 0,
    Branch VARCHAR(50),

    FOREIGN KEY (Customer_ID)
    REFERENCES Customer(Customer_ID)
);

```

Output: -

✓	4	14:28:07	CREATE TABLE Account (Account_No INT PRI...	0 row(s) affected	0.063 sec
---	---	----------	--	-------------------	-----------

Query 4: - CREATE TRANSACTION TABLE

```

//
CREATE TABLE Bank_Transaction (
    Transaction_ID INT PRIMARY KEY AUTO_INCREMENT,
    Account_No INT,
    Transaction_Type VARCHAR(20),
    Amount DECIMAL(12,2),
    Transaction_Date DATETIME DEFAULT CURRENT_TIMESTAMP,

    FOREIGN KEY (Account_No)
    REFERENCES Account(Account_No)
);

```

Output: -

✓	5	14:29:36	CREATE TABLE Bank_Transaction (Transaction...	0 row(s) affected	0.078 sec
---	---	----------	--	-------------------	-----------

Query 5: - CREATE LOAN TABLE

```

0 • CREATE TABLE Loan (
1     Loan_ID INT PRIMARY KEY,
2     Customer_ID INT,
3     Loan_Type VARCHAR(30),
4     Loan_Amount DECIMAL(12,2),
5     Interest_Rate DECIMAL(5,2),
6
7     FOREIGN KEY (Customer_ID)
8     REFERENCES Customer(Customer_ID)
9 );

```

Output: -

✓	6	14:30:49	CREATE TABLE Loan (Loan_ID INT PRIMARY KEY, Customer_ID INT, Loan_Type VARCHAR(30), Loan_Amount DECIMAL(12,2), Interest_Rate DECIMAL(5,2), FOREIGN KEY (Customer_ID) REFERENCES Customer(Customer_ID))	0 row(s) affected	0.031 sec
---	---	----------	---	-------------------	-----------

Query 6: - INSERT CUSTOMER DATA

```

INSERT INTO Customers
(Customer_ID, Customer_Name, Phone, Email, City)
VALUES
(101, 'Raghuveer', '9876543210', 'raghuveer@gmail.com', 'Hyderabad'),
(102, 'Durga prasad', '9876543211', 'durgaprasad@gmail.com', 'Vijayawada'),
(103, 'Vishnu', '9876543212', 'vishnu@gmail.com', 'Bangalore'),
(104, 'Varshith Reddy', '9876543213', 'varshith@gmail.com', 'Chennai'),
(105, 'Ananya Varshitha', '9876543214', 'ananya@gmail.com', 'Hyderabad');

```

Output: -

✓	7	14:33:18	INSERT INTO Customer (Customer_ID, Customer_Name, Phone, Email, City) VALUES (101, 'Aswitha Patel', '9876543210', 'aswitha@gmail.com', 'Hyderabad'), (102, 'Durga prasad', '9876543211', 'durgaprasad@gmail.com', 'Vijayawada'), (103, 'Vishnu', '9876543212', 'vishnu@gmail.com', 'Bangalore'), (104, 'Varshith Reddy', '9876543213', 'varshith@gmail.com', 'Chennai'), (105, 'Ananya Varshitha', '9876543214', 'ananya@gmail.com', 'Hyderabad')	5 row(s) affected	Records: 5 Duplicates: 0 Warnings: 0
---	---	----------	---	-------------------	--------------------------------------

Query 7: - INSERT ACCOUNT DATA

```

INSERT INTO Account
(Account_No, Customer_ID, Account_Type, Balance, Branch)
VALUES
(10001, 101, 'Savings', 50000, 'Hyderabad'),
(10002, 102, 'Savings', 75000, 'Vijayawada'),
(10003, 103, 'Current', 120000, 'Bangalore'),
(10004, 104, 'Savings', 45000, 'Chennai'),
(10005, 105, 'Current', 90000, 'Hyderabad');

```

Output: -

```
✓ 16 14:40:14 INSERT INTO Account (Account_No, Customer_ID, Amount) 5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0 0.031 sec
```

Query 8:- INSERT TRANSACTION DATA

```
INSERT INTO Bank_Transaction
(Account_No, Transaction_Type, Amount)
VALUES
(10001, 'DEPOSIT', 10000),
(10002, 'DEPOSIT', 15000),
(10003, 'WITHDRAW', 20000),
(10004, 'DEPOSIT', 5000),
(10005, 'WITHDRAW', 10000);
```

Output: -

```
✓ 17 14:41:48 INSERT INTO Bank_Transaction (Account_No, Transaction_Type, Amount) 5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0 0.031 sec
```

Query 9: - INSERT LOAN DATA

```
INSERT INTO Loan
(Loan_ID, Customer_ID, Loan_Type, Loan_Amount, Interest_Rate)
VALUES
(501, 101, 'Home Loan', 5000000, 7.5),
(502, 102, 'Education Loan', 1000000, 6.5),
(503, 103, 'Car Loan', 800000, 8.2),
(504, 104, 'Personal Loan', 500000, 10.5);
```

Output: -

```
✓ 18 14:43:11 INSERT INTO Loan (Loan_ID, Customer_ID, Loan_Type, Loan_Amount, Interest_Rate) 4 row(s) affected Records: 4 Duplicates: 0 Warnings: 0 0.047 sec
```

Query 10: - DISPLAY TABLE DATA

```
SELECT * FROM Customers;
```

Output: -

	Customer_ID	Customer Name	Phone	Email	City
▶	101	Vidhath	9876543210	vidhath@gmail.com	Hyderabad
	102	Raghuveer	9876543211	raghuveer@gmail.com	Vijayawada
	103	Durga prasad	9876543212	durgaprasad@gmail.com	Bangalore
	104	Vishnu	9876543213	vishnu@gmail.com	Chennai
	105	Varshith Reddy	9876543214	varshith@gmail.com	Hyderabad
•	NULL	NULL	NULL	NULL	NULL

SELECT * FROM Account;

	Account_No	Customer_ID	Account_Type	Balance	Branch
▶	10001	101	Savings	50000.00	Hyderabad
	10002	102	Savings	75000.00	Vijayawada
	10003	103	Current	120000.00	Bangalore
	10004	104	Savings	45000.00	Chennai
	10005	105	Current	90000.00	Hyderabad
•	NULL	NULL	NULL	NULL	NULL

SELECT * FROM Bank_Transaction;

	Transaction_ID	Account_No	Transaction_Type	Amount	Transaction_Date
▶	1	10001	DEPOSIT	10000.00	2026-08-15 14:41:48
	2	10002	DEPOSIT	15000.00	2026-08-15 14:41:48
	3	10003	WITHDRAW	20000.00	2026-08-15 14:41:48
	4	10004	DEPOSIT	5000.00	2026-08-15 14:41:48
	5	10005	WITHDRAW	10000.00	2026-08-15 14:41:48
•	NULL	NULL	NULL	NULL	NULL

SELECT * FROM Loan;

	Loan_ID	Customer_ID	Loan_Type	Loan_Amount	Interest_Rate
▶	501	101	Home Loan	5000000.00	7.50
	502	102	Education Loan	1000000.00	6.50
	503	103	Car Loan	800000.00	8.20
	504	104	Personal Loan	500000.00	10.50
•	NULL	NULL	NULL	NULL	NULL

Query 11: - PROCEDURE TO FIND CUSTOMER ACCOUNTS


```

DELIMITER //
CREATE PROCEDURE GetCustomerAccounts(
    IN p_Customer_ID INT
)
BEGIN
    SELECT
        C.Customer_ID,
        C.Customer_Name,
        A.Account_No,
        A.Account_Type,
        A.Balance,
        A.Branch
    FROM Customer C
    JOIN Account A
        ON C.Customer_ID = A.Customer_ID
    WHERE C.Customer_ID = p_Customer_ID;
END //
DELIMITER ;
CALL GetCustomerAccounts(101);

```

Output: -

```

✓ 24 14:52:47 CREATE PROCEDURE GetCustomerAccounts( IN p_Customer_ID INT ) BEGIN SELECT C.Cust... 0 row(s) affected
✓ 25 14:52:51 CALL GetCustomerAccounts(101) 1 row(s) returned

```

	Customer_ID	Customer Name	Account_No	Account_Type	Balance	Branch
▶	101	Vidhath	10001	Savings	50000.00	Hyderabad

Query 12: - PROCEDURE TO DEPOSIT MONEY

```
DELIMITER //
```

```
CREATE PROCEDURE DepositMoney(  
    IN p_Account_No INT,  
    IN p_Amount DECIMAL(12,2)  
)  
BEGIN  
    UPDATE Account  
    SET Balance = Balance + p_Amount  
    WHERE Account_No = p_Account_No;  
END //  
DELIMITER ;  
CALL DepositMoney(10001, 5000);
```

Output: -

✓	25	14:55:51	CALL GetCustomerAccounts(10001)	1 row(s) returned
✓	26	14:55:53	CREATE PROCEDURE DepositMoney(IN p_A...	0 row(s) affected
✓	27	14:56:10	CALL DepositMoney(10001, 5000)	1 row(s) affected

Query 13: - PROCEDURE TO WITHDRAW MONEY

```
DELIMITER //
```

```
CREATE PROCEDURE WithdrawMoney(  
    IN p_Account_No INT,  
    IN p_Amount DECIMAL(12,2)  
)  
  
BEGIN  
  
    UPDATE Account  
    SET Balance = Balance - p_Amount  
    WHERE Account_No = p_Account_No;  
  
END //
```

```
DELIMITER ;
```

```
CALL WithdrawMoney(10001, 3000);
```

Output: -

✓	28	14:57:56	CREATE PROCEDURE WithdrawMoney(IN p...	0 row(s) affected
✓	29	14:58:00	CALL WithdrawMoney(10001, 3000)	1 row(s) affected

Query 14: - TRIGGER – PREVENT NEGATIVE DEPOSIT


```
DELIMITER //
```

```
• CREATE TRIGGER CheckTransactionAmount
  BEFORE INSERT ON Bank_Transaction
  FOR EACH ROW
  BEGIN
    IF NEW.Amount <= 0 THEN
      SIGNAL SQLSTATE '45000'
      SET MESSAGE_TEXT =
        'Transaction amount must be greater than zero';
    END IF;
  END //
```

```
DELIMITER ;
```

Output: -

✓	30	14:59:57	CREATE TRIGGER CheckTransactionAmount B...	0 row(s) affected
---	----	----------	--	-------------------

Query 15: - CREATE TRANSACTION AUDIT TABLE

```
CREATE TABLE Transaction_Audit (
  Audit_ID INT PRIMARY KEY AUTO_INCREMENT,
  Transaction_ID INT,
  Account_No INT,
  Transaction_Type VARCHAR(20),
  Amount DECIMAL(12,2),
  Audit_Date DATETIME DEFAULT CURRENT_TIMESTAMP
);
```

Output: -

✓	31	15:01:02	CREATE TABLE Transaction_Audit (Audit_ID ...	0 row(s) affected	0.062 sec
---	----	----------	---	-------------------	-----------

Query 16: - TRIGGER – TRANSACTION AUDIT

```
DELIMITER //
```

- ```
CREATE TRIGGER TransactionAudit
AFTER INSERT ON Bank_Transaction
FOR EACH ROW
BEGIN
 INSERT INTO Transaction_Audit
 (
 Transaction_ID,
 Account_No,
 Transaction_Type,
 Amount
)
 VALUES
 (
 NEW.Transaction_ID,
 NEW.Account_No,
 NEW.Transaction_Type,
 NEW.Amount
);
END //
```

```
DELIMITER ;
```

Output: -

|   |             |                                               |                   |           |
|---|-------------|-----------------------------------------------|-------------------|-----------|
| ✓ | 31 15:01:02 | CREATE TABLE Transaction_Audit ( Audit_ID ... | 0 row(s) affected | 0.062 sec |
|---|-------------|-----------------------------------------------|-------------------|-----------|

Test :-

```
2520030597-Vidhath
(Account_No, Transaction_Type, Amount)
VALUES
(10001, 'DEPOSIT', 2500);
```

Check: -

|   |             |                                               |                   |           |
|---|-------------|-----------------------------------------------|-------------------|-----------|
| ✓ | 32 15:04:10 | INSERT INTO Bank_Transaction (Account_No, ... | 1 row(s) affected | 0.031 sec |
|---|-------------|-----------------------------------------------|-------------------|-----------|

Query 17: - PROCEDURE – ACCOUNT TRANSFER

DELIMITER //

```
CREATE PROCEDURE TransferMoney(
 IN SenderAccount INT,
 IN ReceiverAccount INT,
 IN TransferAmount DECIMAL(12,2)
)

BEGIN

 DECLARE SenderBalance DECIMAL(12,2);

 SELECT Balance
 INTO SenderBalance
 FROM Account
 WHERE Account_No = SenderAccount;

 IF SenderBalance < TransferAmount THEN

 SIGNAL SQLSTATE '45000'
 SET MESSAGE_TEXT =
 'Transfer failed: Insufficient balance';
 END IF;
END
```

```

 IF TransferFailed = 'Insufficient balance' THEN
 -- Transfer failed: insufficient balance

 ELSE

 UPDATE Account
 SET Balance = Balance - TransferAmount
 WHERE Account_No = SenderAccount;

 UPDATE Account
 SET Balance = Balance + TransferAmount
 WHERE Account_No = ReceiverAccount;

 END IF;

 END //

DELIMITER ;
CALL TransferMoney(10001, 10002, 5000);

```

Output: -

|   |    |          |                                          |                   |           |
|---|----|----------|------------------------------------------|-------------------|-----------|
| ✓ | 34 | 15:08:19 | CREATE PROCEDURE TransferMoney( IN Se... | 0 row(s) affected | 0.04      |
| ✓ | 35 | 15:08:53 | CALL TransferMoney(10001, 10002, 5000)   | 1 row(s) affected | 0.031 sec |

CHECK

```

SELECT *
FROM Account
WHERE Account_No IN (10001,10002);

```

Output: -

|   | Account_No | Customer_ID | Account_Type | Balance  | Branch     |
|---|------------|-------------|--------------|----------|------------|
| ▶ | 10001      | 101         | Savings      | 47000.00 | Hyderabad  |
|   | 10002      | 102         | Savings      | 80000.00 | Vijayawada |
| * | NULL       | NULL        | NULL         | NULL     | NULL       |

Query 18: - PROCEDURE WITH IN, OUT PARAMETERS

DELIMITER //

```
CREATE PROCEDURE GetBalance(
 IN p_Account_No INT,
 OUT p_Balance DECIMAL(12,2)
)

BEGIN

 SELECT Balance
 INTO p_Balance
 FROM Account
 WHERE Account_No = p_Account_No;

END //
```

DELIMITER ;

```
CALL GetBalance(10001, @CurrentBalance);
SELECT @CurrentBalance;
```

Output: -

|   |    |          |                                          |                   |
|---|----|----------|------------------------------------------|-------------------|
| ✓ | 37 | 15:11:44 | CREATE PROCEDURE GetBalance( IN p_Acc... | 0 row(s) affected |
| ✓ | 38 | 15:11:48 | CALL GetBalance(10001, @CurrentBalance)  | 1 row(s) affected |

| Result Grid     |          | Filter R |
|-----------------|----------|----------|
| @CurrentBalance |          |          |
| ▶               | 47000.00 |          |

Query 19: -TRIGGER COMMANDS

```
SHOW TRIGGERS;
```

Output: -

| Result Grid              |        | Filter Rows      | Export: CSV                          | Wrap Cell Content: ON |                        |                                          |  |  |
|--------------------------|--------|------------------|--------------------------------------|-----------------------|------------------------|------------------------------------------|--|--|
| Trigger                  | Event  | Table            | Statement                            | Timing                | Created                | sql_mode                                 |  |  |
| ▶ CheckTransactionAmount | INSERT | bank_transaction | BEGIN IF NEW.Amount <= 0 THEN SIG... | BEFORE                | 2026-08-15 14:59:57.20 | ONLY_FULL_GROUP_BY,STRICT_TRANS_TABLE... |  |  |

Query 20: - PROCEDURE MANAGEMENT COMMANDS

```

SHOW PROCEDURE STATUS
WHERE Db = 'BankDB';
DROP PROCEDURE GetAllCustomers;

```

Output: -

|   | Db     | Name                | Type      | Definer        | Modified            | Created             | Security_type | Comment | character_set_client | collation_connection | Databa: Collation |
|---|--------|---------------------|-----------|----------------|---------------------|---------------------|---------------|---------|----------------------|----------------------|-------------------|
| ▶ | bankdb | DepositMoney        | PROCEDURE | root@localhost | 2026-08-15 14:55:53 | 2026-08-15 14:55:53 | DEFINER       |         | utf8mb4              | utf8mb4_0900_ai_ci   | utf8mb4           |
|   | bankdb | GetBalance          | PROCEDURE | root@localhost | 2026-08-15 15:11:44 | 2026-08-15 15:11:44 | DEFINER       |         | utf8mb4              | utf8mb4_0900_ai_ci   | utf8mb4           |
|   | bankdb | GetCustomerAccounts | PROCEDURE | root@localhost | 2026-08-15 14:52:47 | 2026-08-15 14:52:47 | DEFINER       |         | utf8mb4              | utf8mb4_0900_ai_ci   | utf8mb4           |
|   | bankdb | TransferMoney       | PROCEDURE | root@localhost | 2026-08-15 15:08:19 | 2026-08-15 15:08:19 | DEFINER       |         | utf8mb4              | utf8mb4_0900_ai_ci   | utf8mb4           |
|   | bankdb | WithdrawMoney       | PROCEDURE | root@localhost | 2026-08-15 14:57:56 | 2026-08-15 14:57:56 | DEFINER       |         | utf8mb4              | utf8mb4_0900_ai_ci   | utf8mb4           |